

## KEE FIELD, WV, USA

WMO: 720160

Lat: 37.601N

Lon: 81.559W

Elev: 544

StdP: 94.96

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 63884

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|-----|------|-----------------------|------|-----|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%  |      |                       |      |     |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |     |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p) |
| 1             | -12.3      | -9.5 | -17.2                         | 0.9 | -7.8 | -14.2 | 1.2 | -5.5 | 4.5                   | 1.6  | 3.8 | 2.6  | 0.3                   | 310  | N/A |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.2                   | 32.2            | 22.6 | 31.0 | 22.1 | 29.9 | 21.8 | 24.8                | 28.9 | 24.0 | 28.3 | 23.3 | 27.3 | 0.8                  | 320  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 23.7                            | 19.8 | 26.5 | 22.8 | 18.7 | 25.7 | 22.3 | 18.1 | 25.2 | 78.6          | 29.4 | 75.2 | 28.4 | 72.3 | 27.3 | 27.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 3.5           | 2.9  | 2.4 | -17.0                      | 33.7 | 4.1                | 1.4 | -20.0                                              | 34.8 | -22.4      | 35.6 | -24.7      | 36.4 | -27.7      | 37.4 |    |    |
| (5)               |      |     | -17.2                      | 25.9 | 4.4                | 1.3 | -20.4                                              | 26.9 | -23.0      | 27.6 | -25.5      | 28.3 | -28.7      | 29.2 |    |    |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 13.0  | 1.3   | 2.8   | 7.4   | 13.2  | 17.8  | 21.6  | 23.2  | 23.0  | 19.9  | 13.7  | 7.6   | 3.8  | (6)  |
| (7)  |                                                                                  | DBStd       | 9.02  | 5.89  | 5.76  | 5.59  | 4.74  | 4.19  | 2.64  | 2.39  | 2.26  | 3.34  | 4.82  | 4.89  | 5.31 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 951   | 274   | 207   | 118   | 24    | 2     | 0     | 0     | 0     | 0     | 21    | 102   | 204  | (8)  |
| (9)  |                                                                                  | HDD18.3     | 2481  | 527   | 434   | 338   | 160   | 63    | 6     | 1     | 1     | 21    | 156   | 323   | 451  | (9)  |
| (10) |                                                                                  | CDD10.0     | 2047  | 6     | 6     | 38    | 121   | 243   | 349   | 409   | 402   | 298   | 134   | 29    | 11   | (10) |
| (11) |                                                                                  | CDD18.3     | 536   | 0     | 0     | 1     | 7     | 46    | 105   | 152   | 144   | 69    | 11    | 0     | 0    | (11) |
| (12) |                                                                                  | CDH23.3     | 4155  | 0     | 1     | 26    | 201   | 434   | 801   | 1079  | 942   | 547   | 119   | 6     | 1    | (12) |
| (13) |                                                                                  | CDH26.7     | 1301  | 0     | 0     | 2     | 46    | 113   | 249   | 374   | 307   | 185   | 24    | 1     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 0.4   | 0.5   | 0.6   | 0.6   | 0.8   | 0.4   | 0.3   | 0.2   | 0.2   | 0.3   | 0.3   | 0.4   | (14) |      |
| (15) | Precipitation                                                                    | PrecAvg     | 1219  | 99    | 83    | 113   | 105   | 125   | 121   | 136   | 104   | 83    | 70    | 79    | 100  | (15) |
| (16) |                                                                                  | PrecMax     | 1664  | 174   | 205   | 204   | 159   | 229   | 214   | 246   | 155   | 169   | 142   | 195   | 172  | (16) |
| (17) |                                                                                  | PrecMin     | 980   | 52    | 27    | 36    | 46    | 50    | 45    | 58    | 39    | 28    | 16    | 22    | 54   | (17) |
| (18) |                                                                                  | PrecStd     | 166   | 32    | 37    | 43    | 35    | 55    | 43    | 47    | 34    | 38    | 35    | 37    | 33   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 19.0  | 20.9  | 26.2  | 30.7  | 31.5  | 32.4  | 33.0  | 32.9  | 32.8  | 29.0  | 23.8  | 21.0 | (19) |
| (20) |                                                                                  |             | MCWB  | 14.6  | 13.3  | 15.2  | 16.2  | 20.9  | 22.3  | 23.8  | 23.8  | 20.9  | 20.5  | 16.4  | 16.3 | (20) |
| (21) |                                                                                  | 2%          | DB    | 15.3  | 17.2  | 22.9  | 27.8  | 29.6  | 31.1  | 32.0  | 31.3  | 31.1  | 26.5  | 20.9  | 17.2 | (21) |
| (22) |                                                                                  |             | MCWB  | 11.8  | 11.4  | 13.4  | 15.6  | 19.9  | 22.4  | 23.3  | 23.0  | 21.0  | 18.3  | 14.1  | 13.8 | (22) |
| (23) |                                                                                  | 5%          | DB    | 12.5  | 14.2  | 20.2  | 26.0  | 27.7  | 29.8  | 30.5  | 30.0  | 28.9  | 24.0  | 17.8  | 14.3 | (23) |
| (24) |                                                                                  |             | MCWB  | 10.1  | 10.1  | 12.3  | 14.9  | 19.6  | 22.0  | 22.8  | 22.7  | 21.1  | 17.3  | 12.7  | 12.2 | (24) |
| (25) |                                                                                  | 10%         | DB    | 10.1  | 11.9  | 17.2  | 22.9  | 26.1  | 27.9  | 28.8  | 28.2  | 26.8  | 21.4  | 15.6  | 12.1 | (25) |
| (26) |                                                                                  |             | MCWB  | 8.3   | 8.5   | 11.4  | 14.0  | 18.8  | 21.1  | 22.2  | 22.3  | 20.0  | 16.4  | 11.7  | 10.7 | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 15.1  | 15.4  | 17.0  | 19.4  | 23.3  | 24.8  | 26.3  | 26.3  | 24.5  | 21.8  | 17.4  | 17.2 | (27) |
| (28) |                                                                                  |             | MADB  | 17.2  | 17.1  | 23.0  | 26.6  | 28.5  | 29.4  | 30.2  | 29.7  | 28.9  | 26.4  | 21.6  | 19.3 | (28) |
| (29) |                                                                                  | 2%          | WB    | 12.6  | 12.7  | 15.5  | 17.9  | 21.8  | 23.9  | 24.9  | 24.7  | 23.1  | 20.5  | 15.5  | 14.8 | (29) |
| (30) |                                                                                  |             | MADB  | 14.4  | 15.5  | 20.6  | 23.6  | 26.9  | 28.5  | 28.9  | 28.7  | 27.5  | 24.3  | 18.6  | 16.6 | (30) |
| (31) |                                                                                  | 5%          | WB    | 10.4  | 10.8  | 14.1  | 16.6  | 20.8  | 23.1  | 24.0  | 23.8  | 22.0  | 18.9  | 14.1  | 12.5 | (31) |
| (32) |                                                                                  |             | MADB  | 12.0  | 13.5  | 18.4  | 21.8  | 25.4  | 27.5  | 28.4  | 27.6  | 26.2  | 21.8  | 16.7  | 13.7 | (32) |
| (33) |                                                                                  | 10%         | WB    | 8.0   | 8.7   | 12.4  | 15.4  | 19.9  | 22.3  | 23.3  | 22.9  | 21.0  | 17.3  | 12.5  | 10.6 | (33) |
| (34) |                                                                                  |             | MADB  | 9.5   | 11.6  | 15.6  | 20.2  | 24.2  | 26.3  | 27.2  | 26.6  | 24.6  | 19.9  | 14.6  | 11.8 | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 10.1  | 10.9  | 12.6  | 14.9  | 13.0  | 12.2  | 11.2  | 12.3  | 12.8  | 12.4  | 9.7   | (35) |      |
| (36) |                                                                                  |             | MADB  | 13.7  | 15.8  | 18.0  | 19.6  | 15.4  | 14.2  | 13.1  | 13.0  | 15.2  | 15.8  | 15.7  | 13.3 | (36) |
| (37) |                                                                                  |             | MCWBR | 9.7   | 10.2  | 9.1   | 8.1   | 6.8   | 6.3   | 5.4   | 5.7   | 6.2   | 7.8   | 9.3   | 9.4  | (37) |
| (38) |                                                                                  | 5% WB       | MADB  | 11.9  | 13.5  | 14.4  | 14.7  | 13.2  | 12.4  | 11.4  | 11.3  | 12.8  | 12.6  | 12.4  | 12.0 | (38) |
| (39) |                                                                                  |             | MCWBR | 9.7   | 10.0  | 8.5   | 7.6   | 6.4   | 6.0   | 5.2   | 5.6   | 6.0   | 7.0   | 8.6   | 9.4  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.300 | 0.310 | 0.333 | 0.375 | 0.418 | 0.443 | 0.472 | 0.481 | 0.377 | 0.347 | 0.314 | 0.293 | (40) |      |
| (41) |                                                                                  | taud        | 2.510 | 2.469 | 2.458 | 2.354 | 2.228 | 2.198 | 2.140 | 2.091 | 2.378 | 2.467 | 2.532 | 2.552 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 893   | 925   | 932   | 905   | 866   | 842   | 813   | 796   | 872   | 870   | 868   | 878   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 78    | 93    | 103   | 121   | 140   | 145   | 152   | 156   | 111   | 92    | 77    | 71    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.95  | 2.59  | 3.72  | 4.89  | 5.47  | 6.02  | 5.62  | 5.26  | 4.45  | 3.33  | 2.43  | 1.72  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.20  | 0.36  | 0.29  | 0.37  | 0.43  | 0.38  | 0.33  | 0.26  | 0.42  | 0.42  | 0.28  | 0.19  | (45) |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (27 neighbors) | +0.43 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +134    | +71     |

Nomenclature: See separate page